

Homework (Habanero)

- Q1.** What type of correlation is shown by the scatter plot where patients' dosage of medicine increases as their blood pressure decreases?
- (A) Positive
(B) Negative
(C) None
(D) Linear
(E) Exponential
- Q2.** A researcher collects data on the growth rate of cells in a controlled environment. Which of the following best describes a line of best fit for this data?
- (A) A line that passes through every data point
(B) A line that minimizes the sum of the squared errors
(C) A line that maximizes the correlation coefficient
(D) A line that is parallel to the x-axis
(E) A line that is perpendicular to the y-axis
- Q3.** What is the purpose of creating a scatter plot in healthcare research?
- (A) To determine the correlation between two variables
(B) To predict the outcome of a treatment
(C) To identify patterns in patient data
(D) To compare the effectiveness of different treatments
(E) All of the above
- Q4.** If a scatter plot shows a strong positive correlation between the amount of medicine administered and the patient's recovery rate, what can be inferred?
- (A) The medicine has a direct impact on the patient's recovery rate
(B) The patient's recovery rate is unrelated to the amount of medicine administered
(C) The data is insufficient to make any conclusions
(D) The correlation is due to an external factor
(E) The medicine is ineffective
- Q5.** A scatter plot shows the relationship between the concentration of a drug and its effectiveness. If the line of best fit has a slope of 2.5, what does this indicate?
- (A) For every 1 unit increase in concentration, the effectiveness increases by 2.5 units
(B) For every 1 unit increase in concentration, the effectiveness decreases by 2.5 units
(C) The concentration has no impact on the effectiveness
(D) The relationship between concentration and effectiveness is exponential
(E) The data is insufficient to make any conclusions
- Q6.** Which of the following is a characteristic of a line of best fit?
- (A) It passes through every data point
(B) It minimizes the sum of the squared errors
(C) It is a horizontal line
(D) It is a vertical line
(E) It is a curve
- Q7.** A researcher collects data on the relationship between the dosage of a vaccine and the resulting immune response. If the scatter plot shows a strong negative correlation, what can be inferred?
- (A) The vaccine is effective in boosting the immune response
(B) The vaccine has no impact on the immune response
(C) The vaccine actually weakens the immune response
(D) The data is insufficient to make any conclusions
(E) The correlation is due to an external factor
- Q8.** What is the purpose of using a line of best fit in medical research?
- (A) To predict patient outcomes based on treatment
(B) To identify patterns in patient data
(C) To compare the effectiveness of different treatments
(D) To determine the correlation between two variables
(E) All of the above

Open Ended Questions (FRQ)

- Q9.** A pharmaceutical company collects data on the relationship between the dosage of a new medicine and the resulting reduction in blood pressure. The data is as follows: (2, 10), (4, 12), (6, 15), (8, 18), (10, 20). Create a scatter plot and draw a line of best fit. Use the line of best fit to predict the reduction in blood pressure for a patient taking a 12-unit dose of the medicine. Show your work and provide a clear explanation of your prediction.
- Q10.** A researcher collects data on the growth rate of bacteria in a controlled environment. The data shows a strong positive correlation between the concentration of a nutrient and the growth rate of the bacteria. Create a scatter plot of the data and draw a line of best fit. Use the line of best fit to predict the growth rate of the bacteria for a nutrient concentration of 15 units. Provide a clear explanation of your prediction and show your work.

Chef's Tips

- Tip 1: Encourage students to use graph paper to create their scatter plots
- Tip 2: Remind students to label their axes and include a title for their graph
- Tip 3: Emphasize the importance of showing work and providing clear explanations for their predictions